# Wireless Interfaces

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**Course Title:** Peripherals & Interfacing
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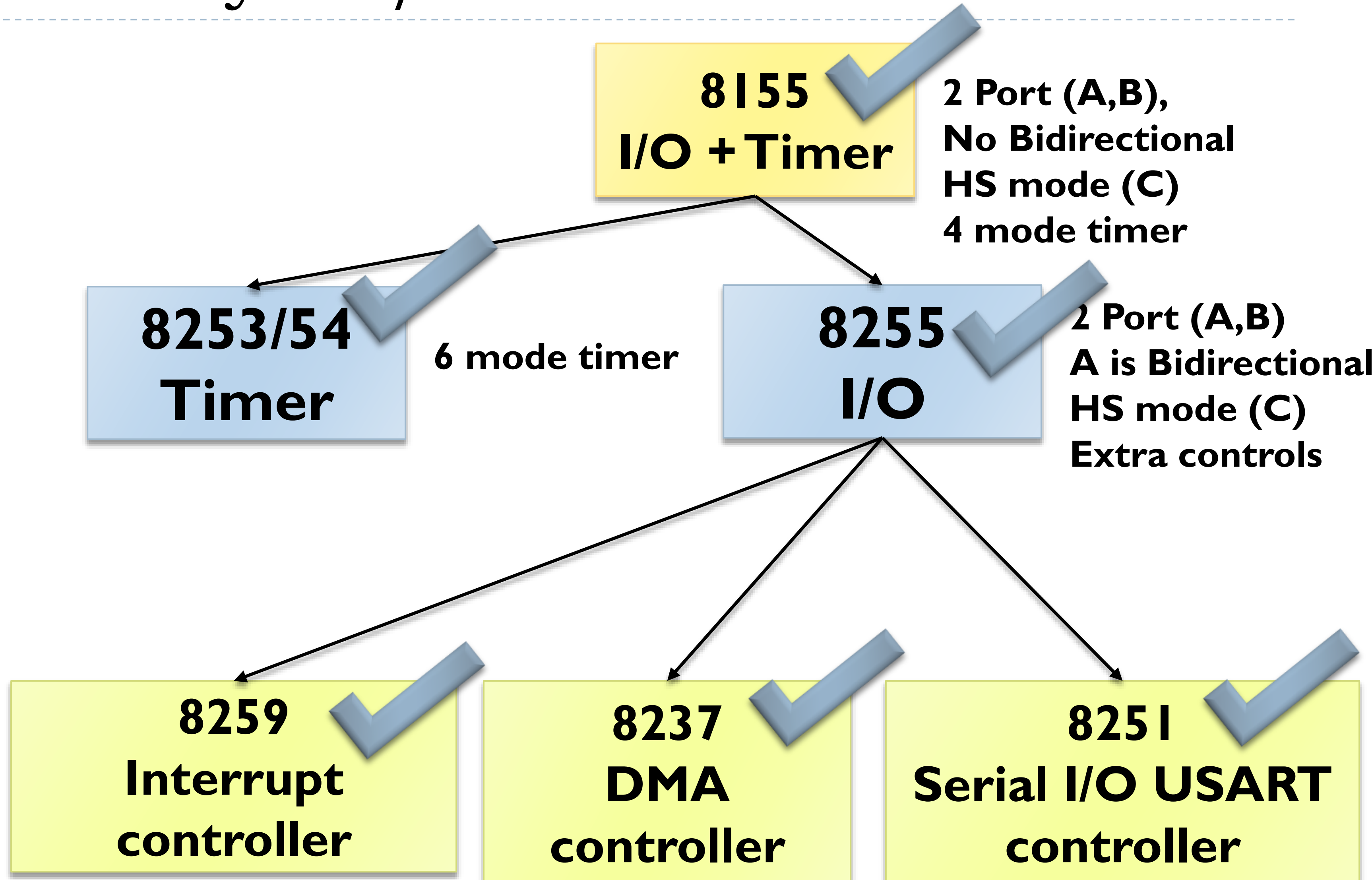
# Lecture References:

- Book:
  - *Microprocessor Architecture, Programming and Applications with 8085 (Chapter-15),* **Author:** Ramesh Gaonkor
  - *Microprocessors and Interfacing: Programming and Hardware,* **Author:** Douglas V. Hall

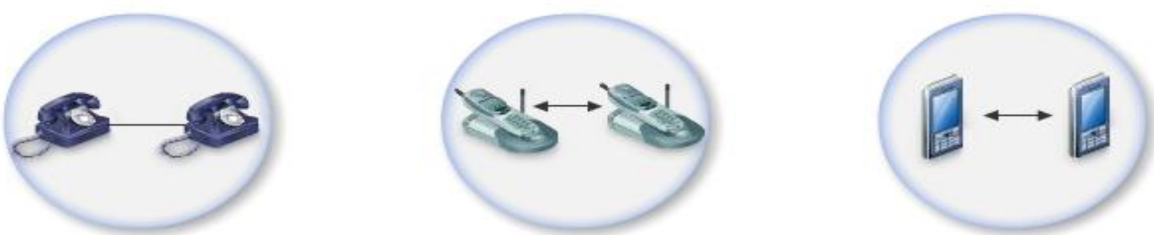
- Lecture Materials:
  - *I/O System Design*, Dr. Esam Al_Qaralleh, CE Department, Princess Sumaya University for Technology.

# Hierarchy of I/O Control Devices



# History of Wireless Communication

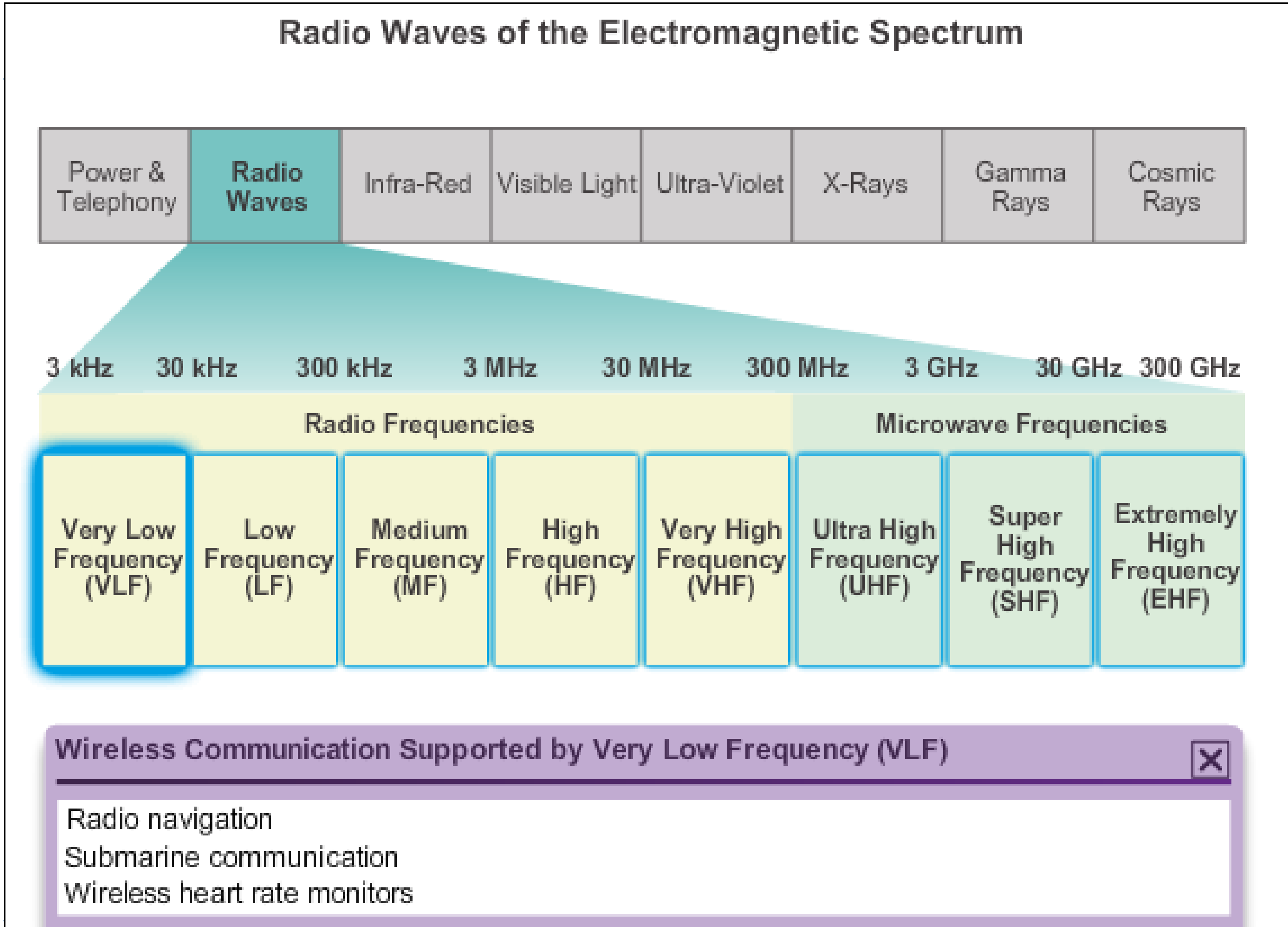
- Guglielmo Marconi invented the wireless telegraph in 1896
  - Communication by encoding alphanumeric characters in analog signal
  - Sent telegraphic signals across the Atlantic Ocean
- 1914 – first voice communication over radio waves
- Communications satellites launched in 1960s
- Advances in wireless technology
  - Radio, television, mobile telephone, communication satellites
- More recently
  - Satellite communications, wireless networking, cellular technology



# Wireless Frequency?

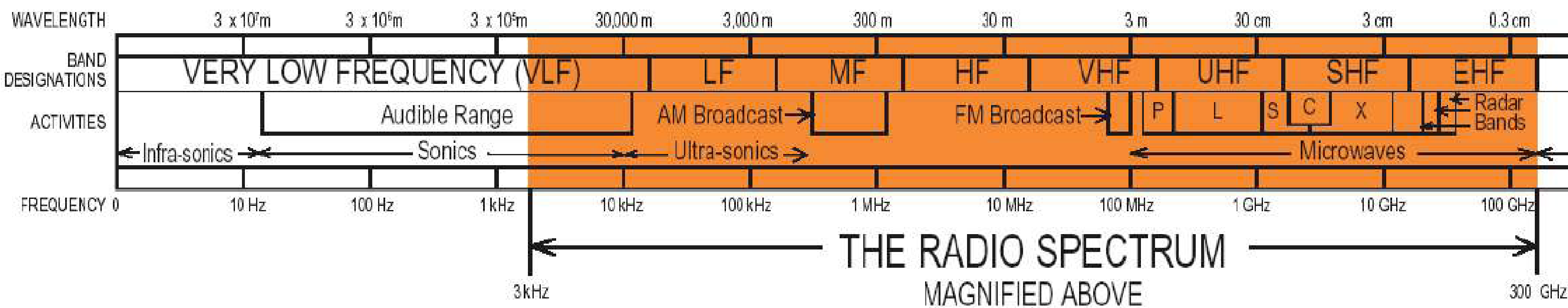
- Transmitting voice and data using electromagnetic waves in open space (atmosphere) or in any medium.
- Electromagnetic waves
  - Travel at speed of light ($c = 3 \times 10^8$ m/s)
  - Has a frequency (f) and wavelength ($\lambda$)
    - $c = f \times \lambda$
  - Higher frequency means higher energy photons
  - The higher the energy photon the more penetrating is the radiation

# Wireless Frequency



# What is RF?

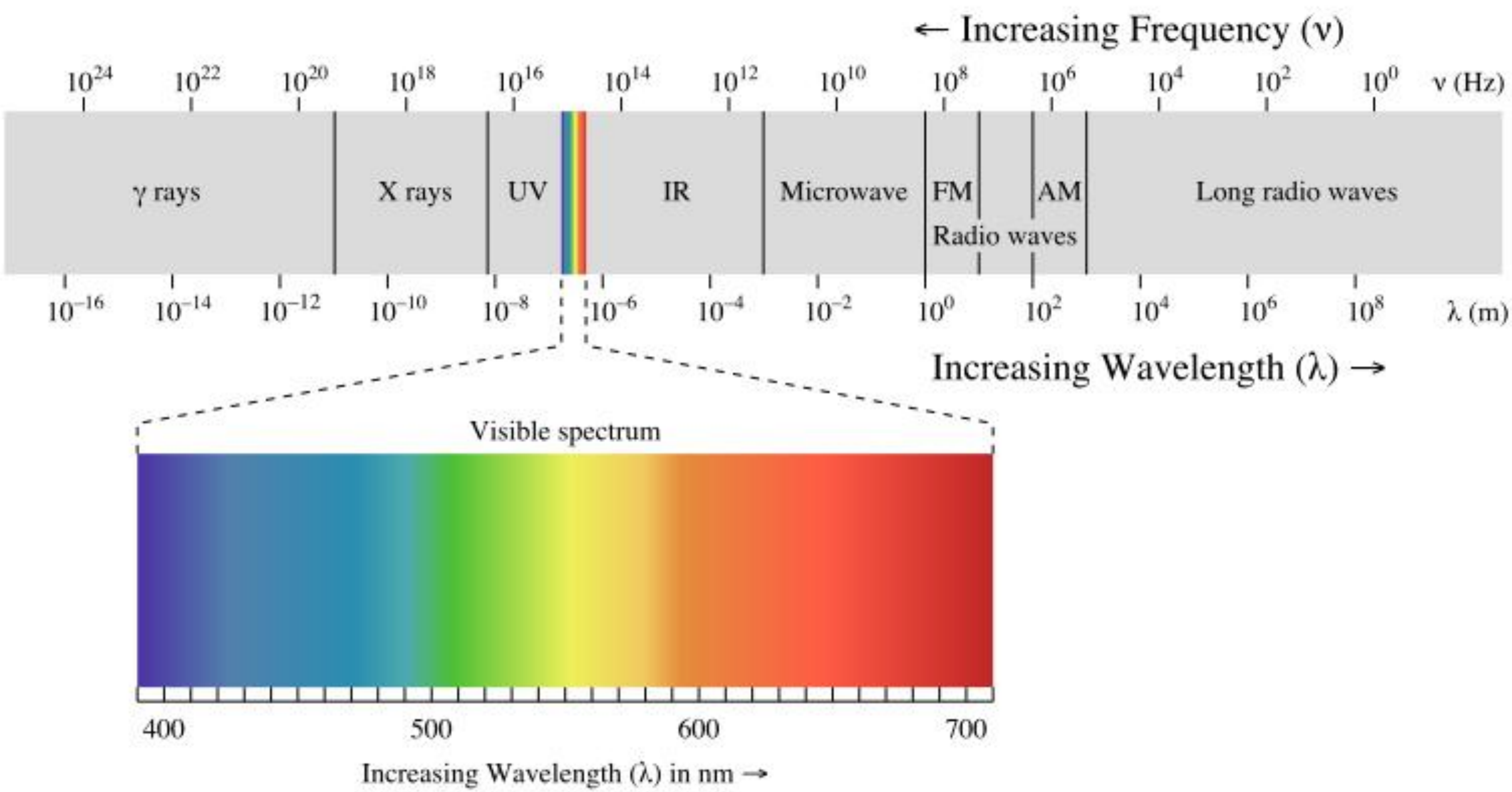
Radio Frequency is an electromagnetic signal with a frequency between 3 kHz and 300 Ghz



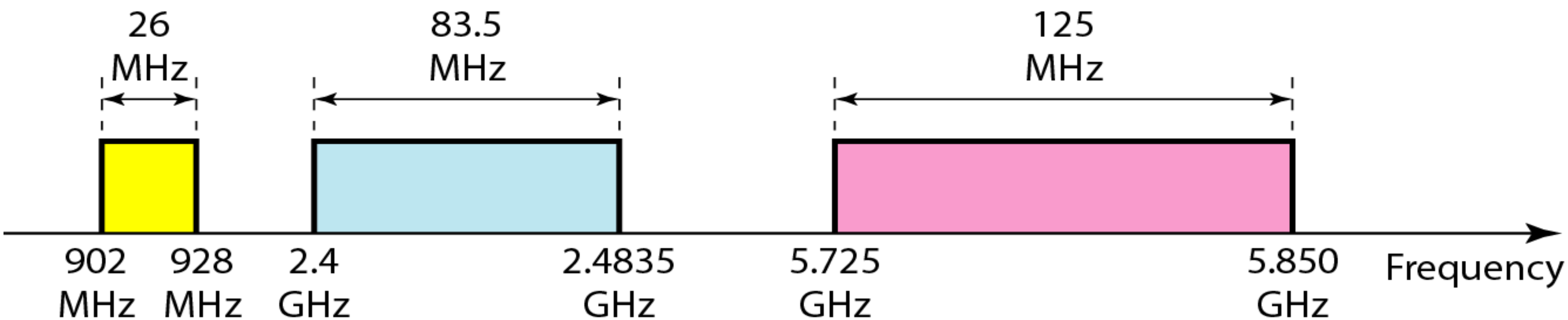

RF signals carry analog or digital information

- Analog: Information content varies continuously over time
    - *Example: radio and TV stations*
- Digital: Information content consists of discrete units (e.g., 0s and 1s)
    - *Example: Cell phones and wireless networks*

# Electromagnetic Radiation Spectrum



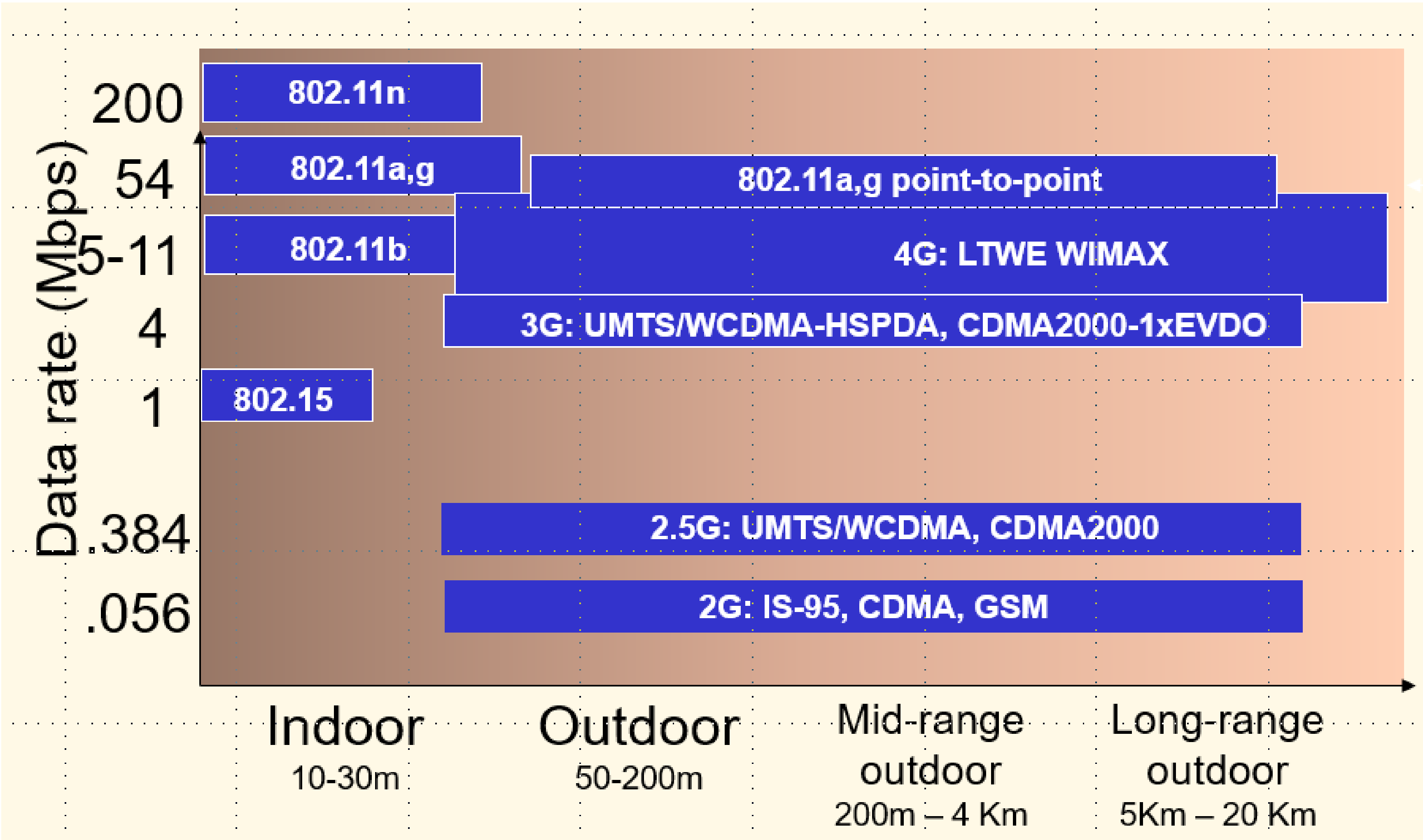
# Industrial, Scientific & Medical (ISM) Band



# Wireless Interfaces

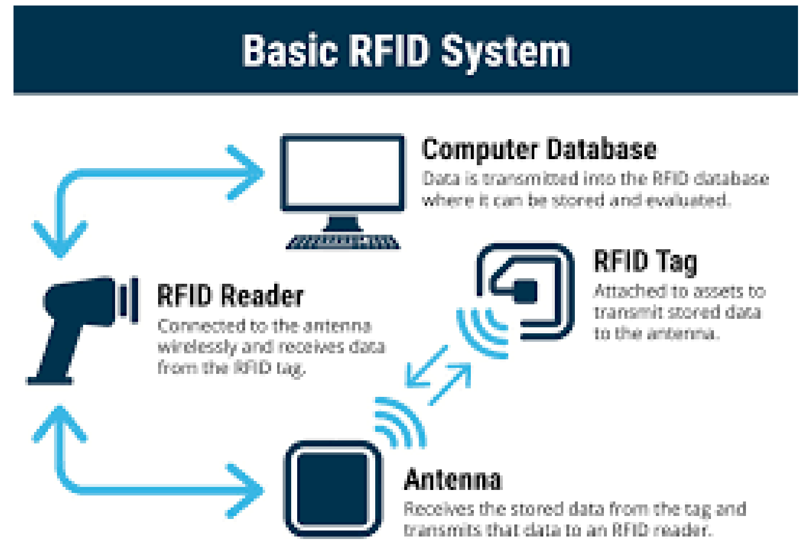
- RFID (Radio Frequency Identifier)
- BT (Bluetooth)
- WiFi (Wireless Fidelity)
- Cellular Technology
- LoRa IoT Interface Standards

# Wireless Link Standards



# RFID Interfaces

- RFID uses radio waves to transfer data from an electronic tag (RFID tag or label), attached to an object, through a reader to identify and track the object.
- The tag's information is stored electronically.
- Some RFID tags can be read from several meters away and beyond the line of sight of the reader.



# RFID Interfaces

- An RFID reader transmits an encoded radio signal to interrogate the tag.

- With a small RF transmitter and receiver, the RFID tag receives the message and responds with its identification information.

- Many RFID tags have no battery. Instead, the tag uses the radio energy transmitted by the reader as its energy source.

# Interfaces: Wireless Protocols

- **IrDA** (Infrared Data Association)
  - IrDA is an international organization
  - Designed to support transmission between two devices over short-range point-to-point infrared.
  - Rate: 9.6 Kb/s – 4 Mb/s
  - Deployed in notebooks, printers, PDAs, cell phones,…
  - MS Windows CE 1.0 the first Windows OS support it
  - Available on several popular embedded OSs

# Bluetooth Interfaces

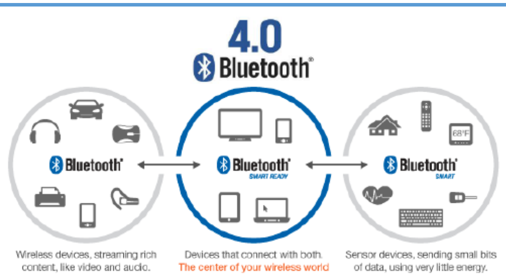
- Bluetooth is a wireless technology which is used for computers, PDAs, mobile phones, headphones, mice and keyboards.
- It transmits very weak signals on a 2.4 to 2.48 gigahertz or 5 gigahertz frequency (ISM Band) for communication over short distances, typically up to 10 metres.

Bluetooth transfer speed up to 1 megabit per second.

# Bluetooth Interfaces

**BT Basic/Enhanced Data Rate**

- 2.45 GHz ISM frequency band
- 79 Channels (USA, Europe) or 23 (Japan) RF channels. 1 MHz channel width
- Frequency Hopping (FH) - Frequency-Hopping/Time-Division-Duplex (FH/TDD) - Adaptive Frequency Hopping
- Devices are organized in piconets
- The master of the piconet provides the clock and the frequency-hopping pattern (ordering of the frequencies)



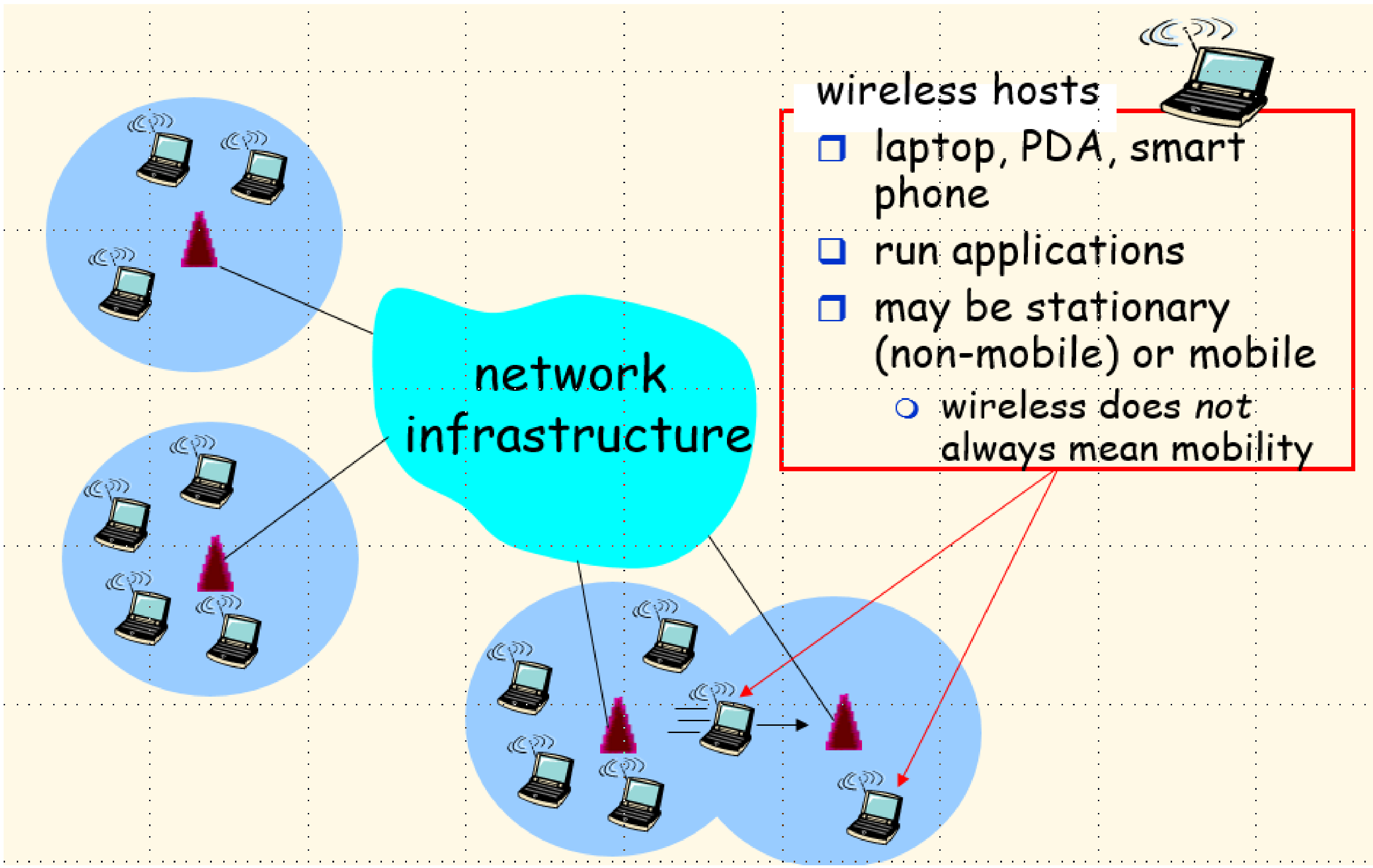

**BT Low Energy**

- 2.45 GHz ISM band
- 40 RF channels. 2 MHz channel width.
- AFH mechanism: FDMA and TDMA
- @FDMA: the 40 physical channels are divided into 3 advertising channels and 37 data channels.
- Devices are organized in piconets.
- The master of the piconet determines the hop interval and the hopping pattern for accessing the 37 physical channels. @TDMA: polling-based data transmission between the master and the slaves of the piconet.

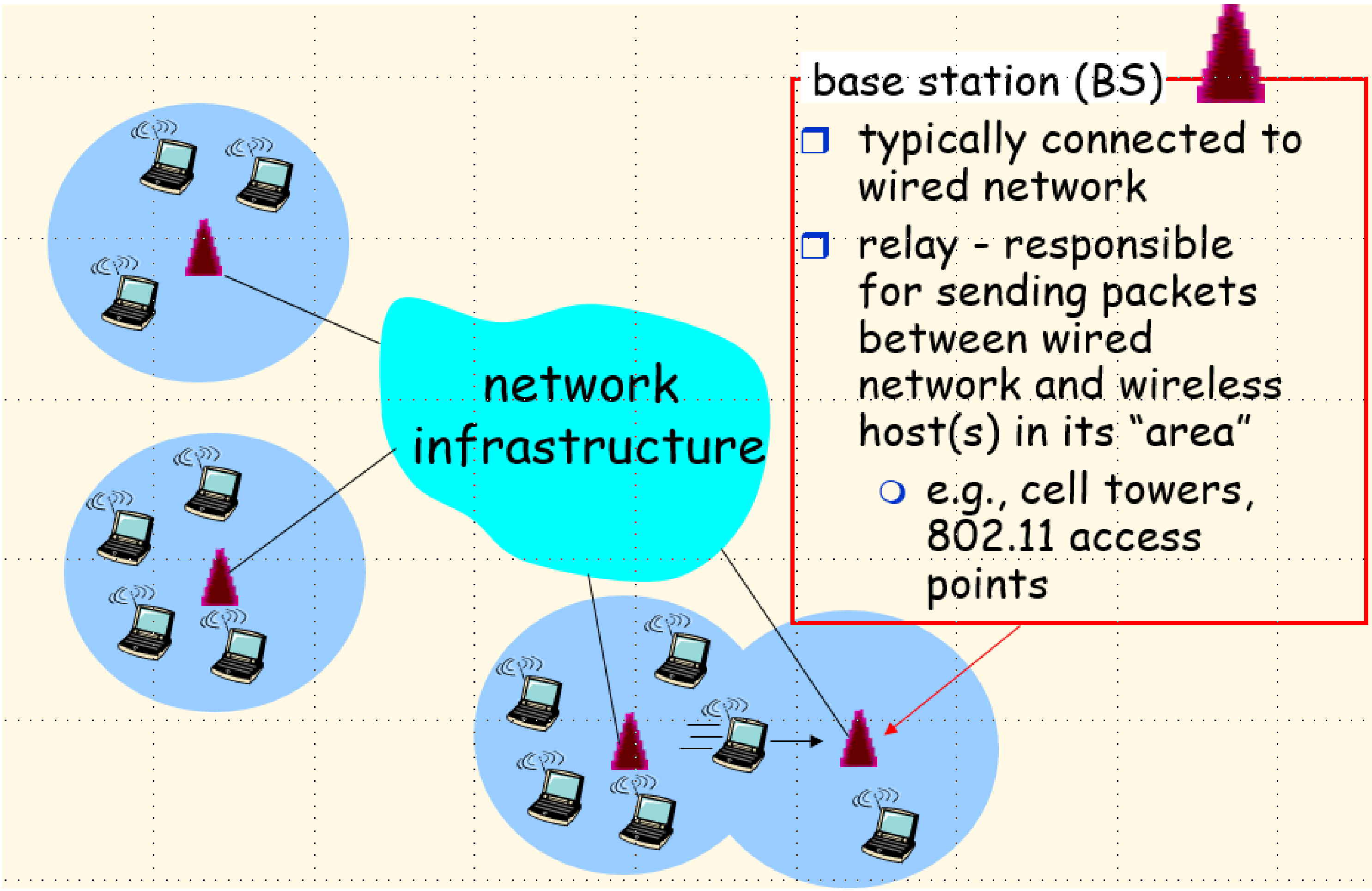
# WiFi Interfaces

- **IEEE 802.11**
  - IEEE proposed standard for WLAN
  - Ad-hoc vs. infrastructure
  - PHY and MAC layers
  - Data rate: 1Mbps, 2Mbps, 11 Mbps, 54 Mbps, 600 Mbps
  - Calls: 2.4 – 2.4835 GHz frequency band (unlicensed band).
  - Use CSMA/CA
  - Signals for transmission: RTS, CTS, and ACK.

# WiFi Interfaces

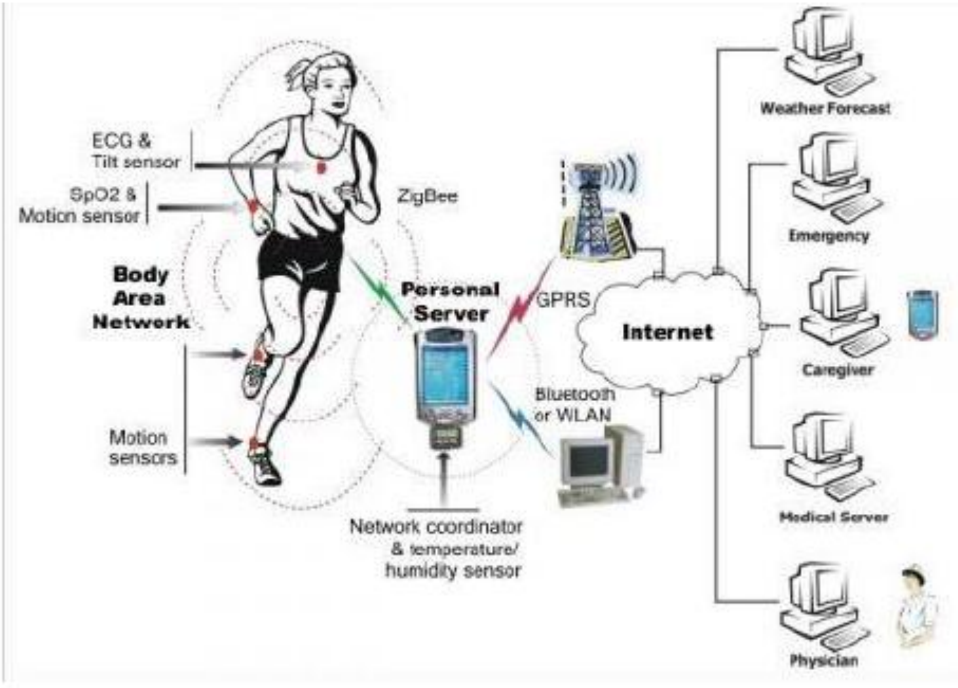


# WiFi Interfaces



# WPAN Interfaces

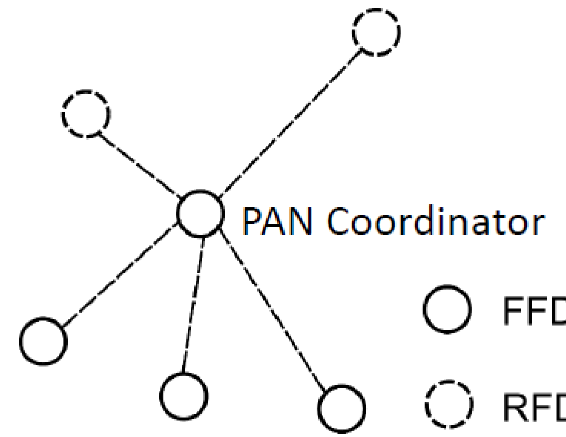
- **IEEE 802.15.4**
  - IEEE proposed standard for WPAN
  - Basically for star-topology
  - PHY and MAC layers
  - Data rate: 250 Kbps
  - Calls: 2.4 – 2.4835 GHz frequency band (unlicensed band).
  - Use CSMA/CA and TDMA



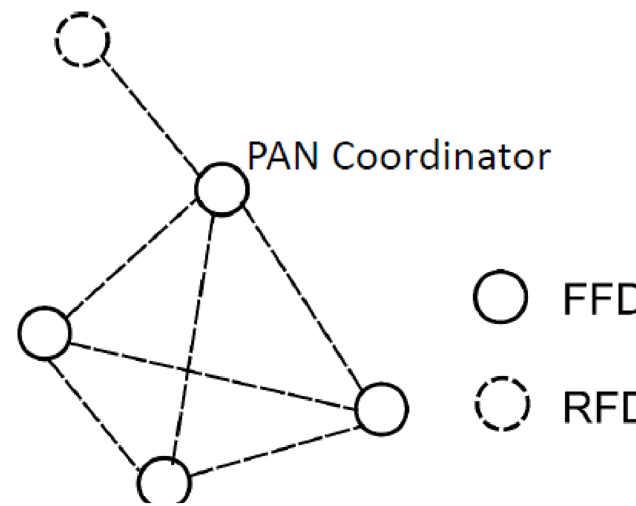
# WPAN Interfaces

Types of logical topologies

- **Star**: each device (FFD or RDF) communicates with the PAN coordinator only
  - Suitable for small-scale networks that operate within a limited space
  - home automation, computer peripherals, peripherals, games, and smart health care




- **Peer-to-peer:** FFD devices can communicate with each other, as long as they are within communication range.
  - More flexible than star, suitable for larger-scale networks that need distributed coordination between peers without the necessity of a central unit.
  - Multi-hopping, Cluster Trees, and Mesh networking



# Embedded IoT Network

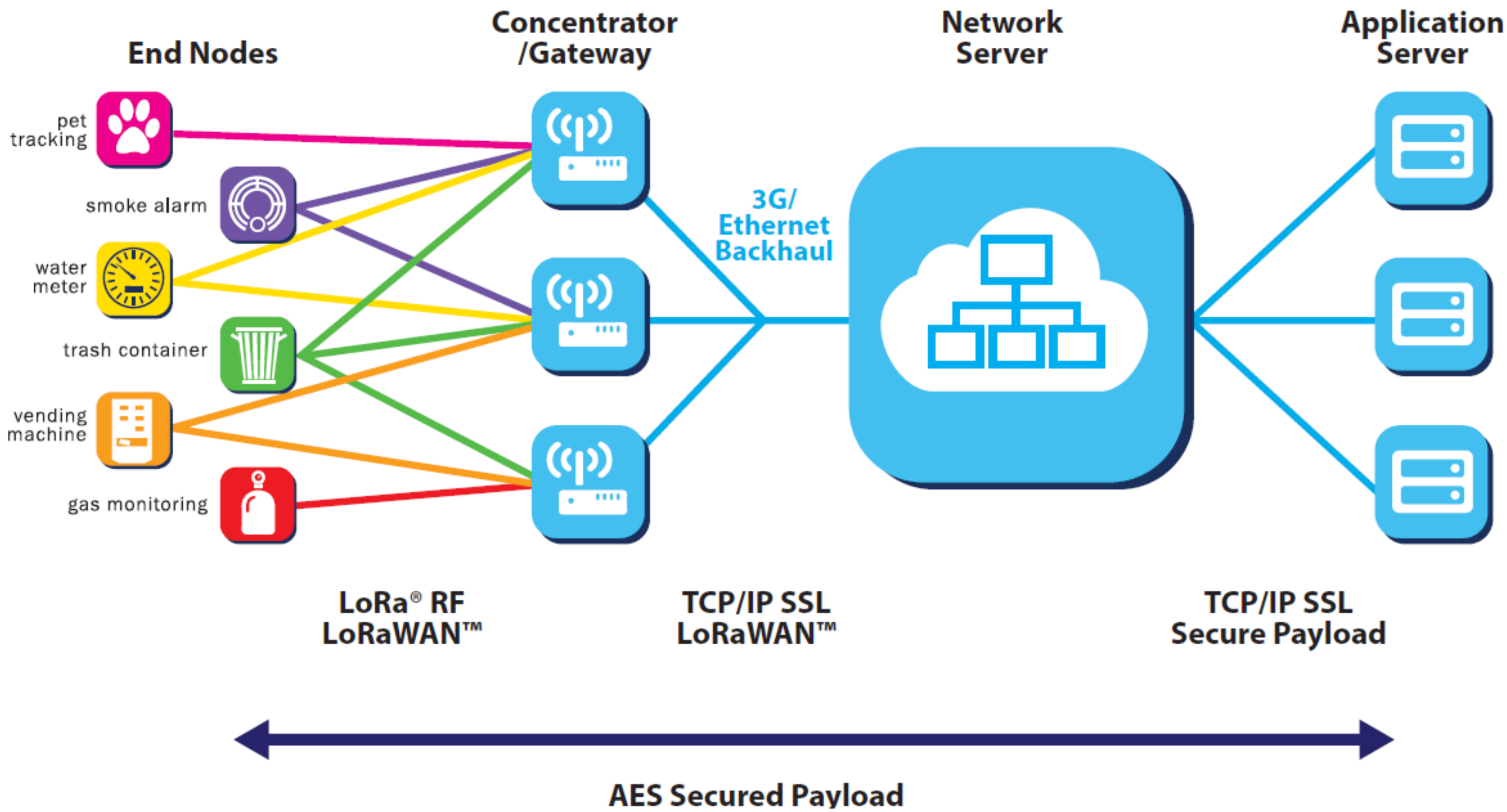
# LoRaWAN: Long Range WAN

- The advantage of LoRa is in the technology's long range capability.
- A single gateway or base station can cover **entire cities or hundreds of square kilometres**.
- Range highly depends on the environment or obstructions in a given location, but LoRa and LoRaWAN have a link budget greater than any other standardized communication technology.
- The link budget, typically given in decibels (dB), is the primary factor in determining the range in a given environment.

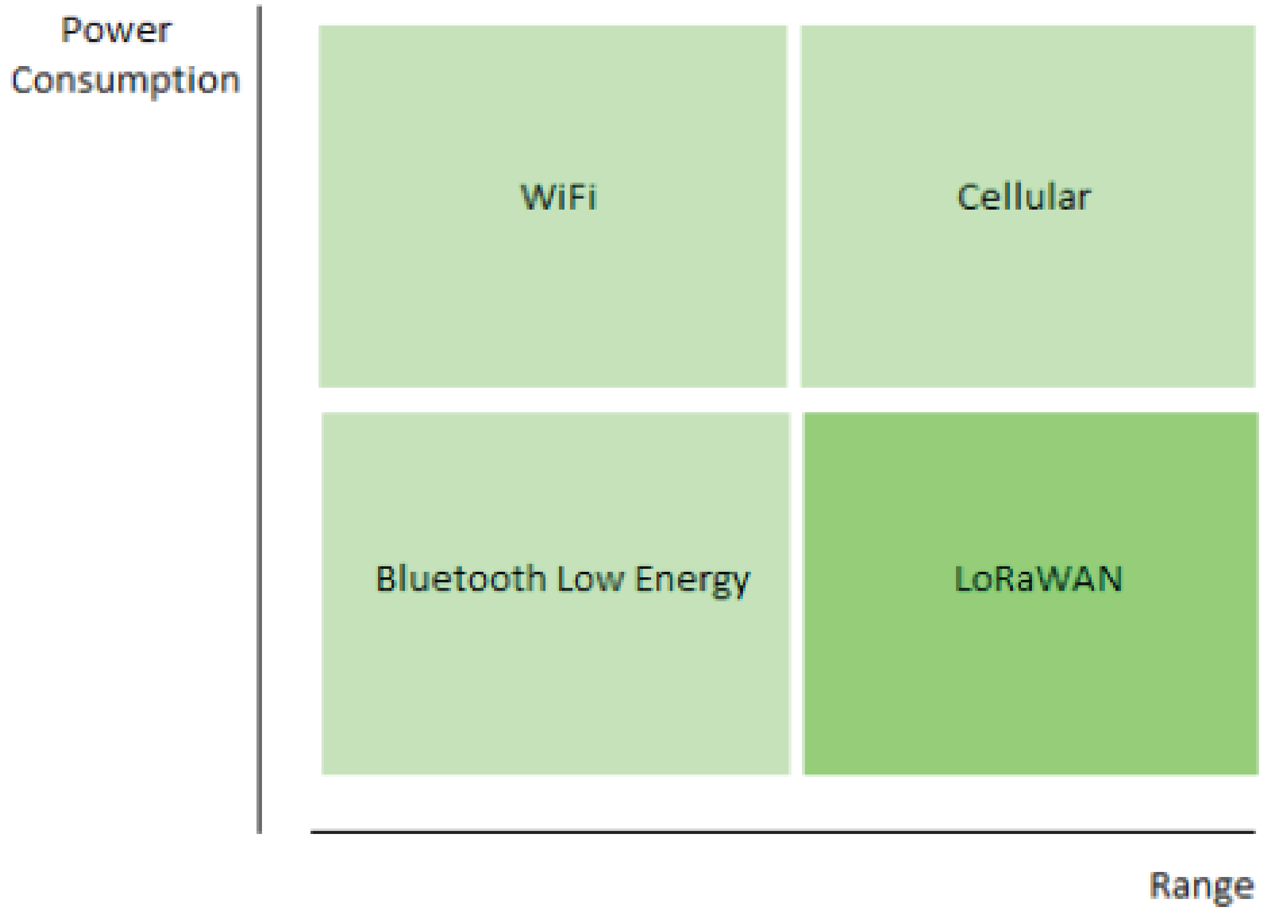
# WHERE DOES LPWAN FIT?

- One technology cannot serve all of the projected applications and volumes for IoT.
- **WiFi and BT** are widely adopted standards and serve the applications related to communicating personal devices quite well.
- **Cellular technology** is a great fit for applications that need high data throughput and have a power source.
- **LPWAN** offers **multi-year battery lifetime** and is designed for sensors and applications that need to send small amounts of data over long distances a few times per hour from varying environments

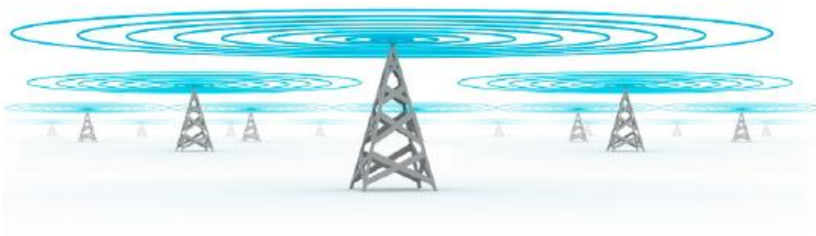
# LoRaWAN Architecture



# Wireless Technology Comparisons for IoT and Embedded Devices



# LoRaWAN: Advantages



**Long Range**

- Greater than cellular
- Deep indoor coverage
- Star topology

**Max Lifetime**

- Low power optimized
- 10-20yr lifetime
- >10x vs cellular M2M

# LoRaWAN: Advantages

**Multi-Usage**

- High capacity
- Multi-tenant
- Public network

**Low Cost**

- Minimal infrastructure
- Low cost end node
- Open SW

# Thank You !!